

# Smart Playlist Generation Using Clustering and Spotify Audio Features

A data-driven approach to automated playlist creation

Dataset: 5,000 Spotify tracks

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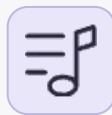
*Xuan Liu*

*Konul Aslanova*





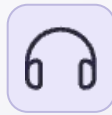
# Project Objective



**Automatically generate playlists grouped by musical similarity**



**Ensure each playlist has between 50 and 250 songs**



**Rely purely on audio features, not metadata**



**Prioritize musical traits that align with human listening perception**

# Our Approach

- **Collect audio features from Spotify**
- **API**
  - **Normalize features using**
  - **MinMaxScaler**
    - **Apply custom feature weights to emphasize perceptual**
    - **relevance**
      - **Perform initial clustering using**
  - **KMeans**
    - **Re-cluster with KMeansConstrained to enforce playlist size**
- **constraints**
  - **Generate final playlists from resulting**
  - **clusters**

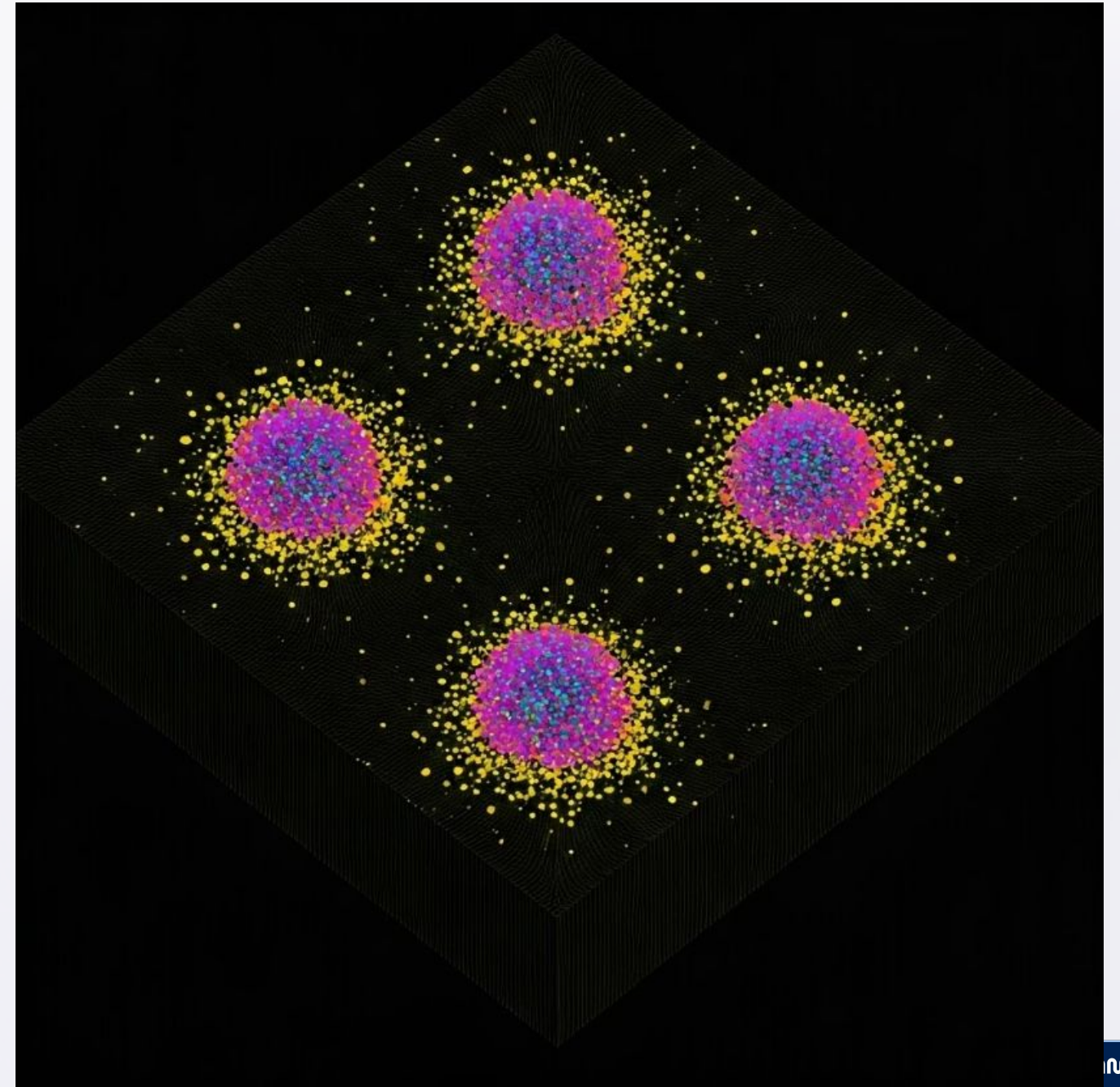
# Initial Clustering with

## KMeans

Chose  $k=5$  based on silhouette and inertia scores then sub-clustering

Good cluster separation, but...

Playlist sizes became inconsistent when sub-clustered

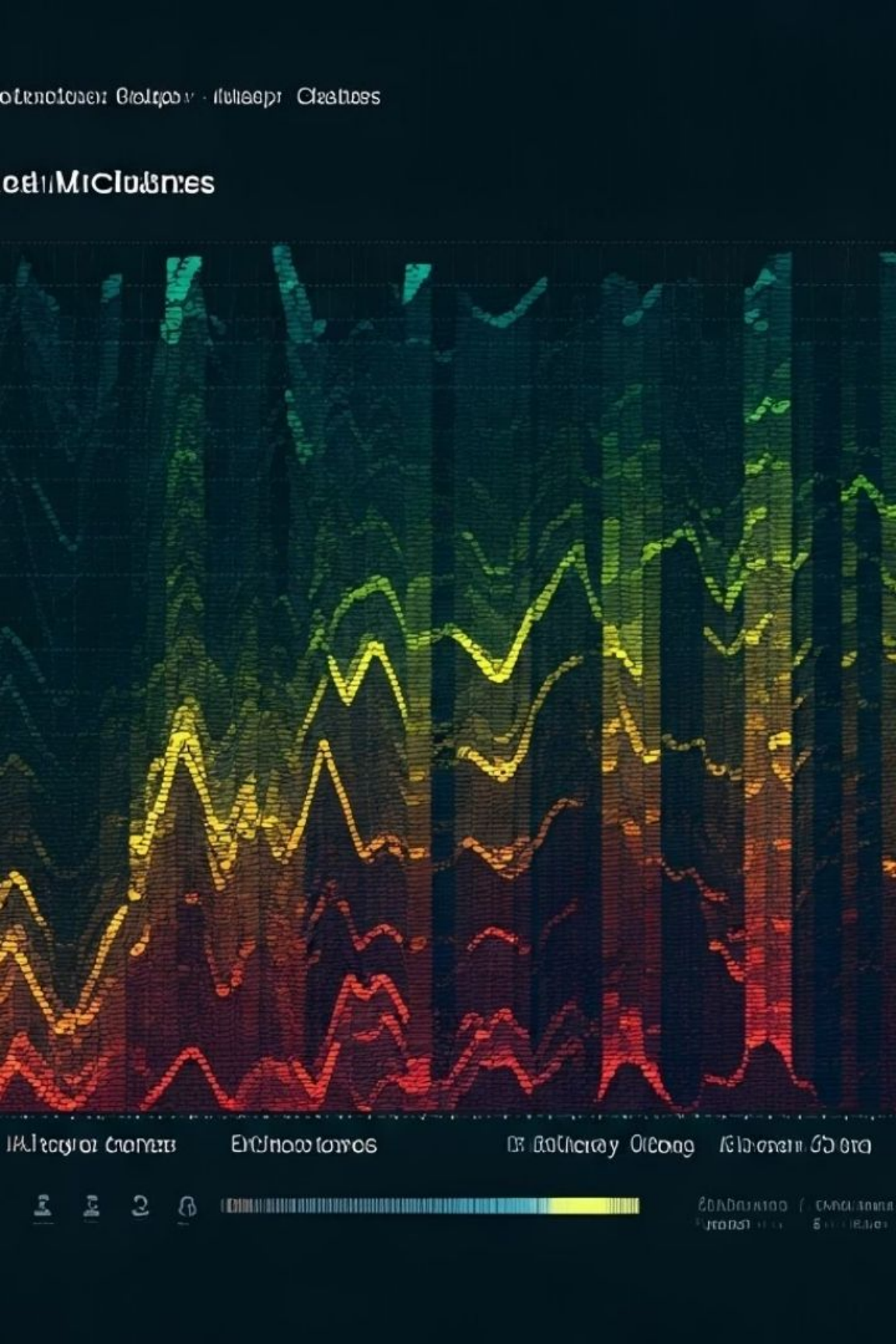




# The Problem



Some clusters had <50 songs, others >250



# Solution: KMeansConstrained + Feature Weighting

## Size Constraints

Supports size\_min and size\_max parameters

## Weighted Features

Re-applied clustering with weighted features to reflect listening experience

## Perceptual Emphasis

- Acousticness & Instrumentalness ( $\times 1.5$ ): distinguish natural vs. electronic and vocal vs. instrumental music
- Danceability ( $\times 1.2$ ): emphasize rhythm
- Duration, Loudness, Speechiness ( $\times 0.5\text{--}0.8$ ): reduced impact due to lower perceptual relevance

## Result

45 balanced, perceptually meaningful playlists

# Features Used

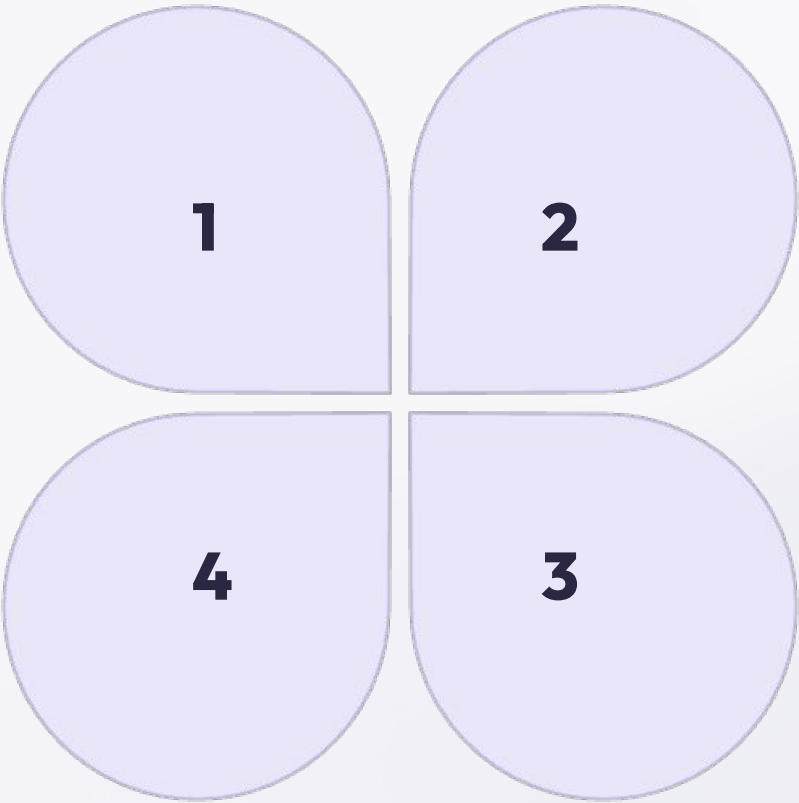
## Weighted Features

(High)

- acousticness (1.5)
- instrumentalness (1.5)
- danceability (1.2)

## Dropped Features

- key, mode, time\_signature (low variance)
- artist, genre, song\_name (non-numeric)



## Weighted Features

(Medium)

- energy (1.0)
- valence (1.0)
- tempo (1.0)

## Weighted Features

(Low)

- speechiness (0.8)
- loudness (0.8)
- liveness (0.7)
- duration\_ms (0.5)



# Playlist Examples



## Playlist 1 (Cluster

**22)** High danceability, mid-tempo, high valence

Feels like a summer chill playlist



## Playlist 2 (Cluster

**14)** Low energy, high acousticness

Ideal for studying or relaxation





# Are Spotify Features Human-Meaningful?



## Mood Matching

Features like energy, tempo, valence often match human-perceived mood



## Style Recognition

Acousticness and instrumentality help surface stylistic similarities



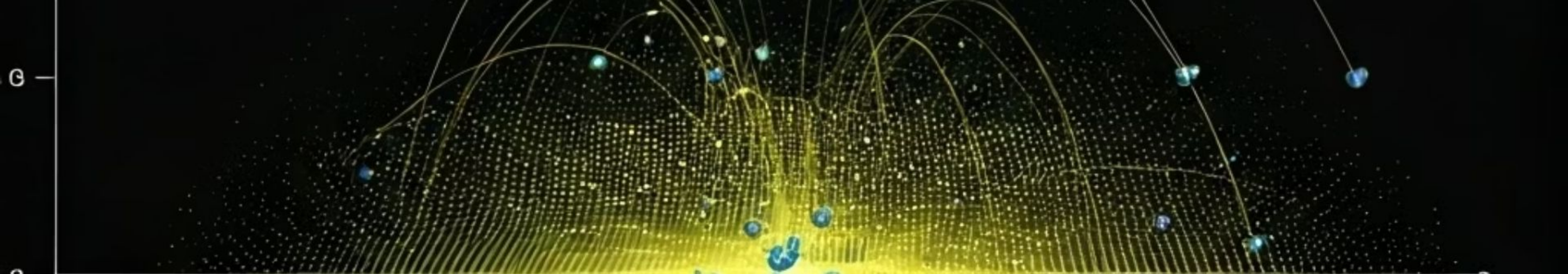
## Missing Context

However, no awareness of lyrics, language, genre or cultural context



## Compensation

Weighting features helps compensate for these gaps



# Is KMeans a Good

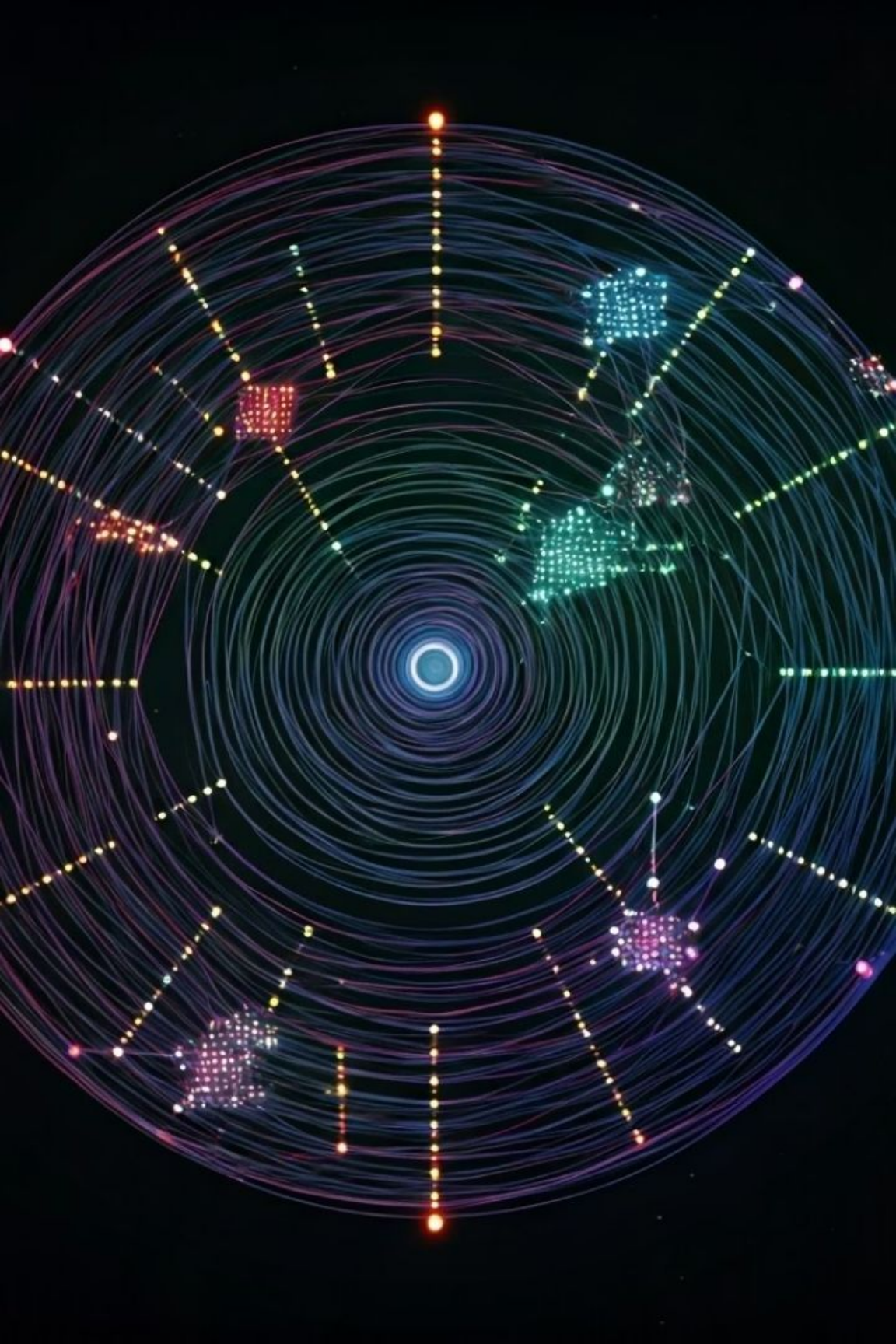
## Fit?

| Pros                                                 | Cons                                                   |
|------------------------------------------------------|--------------------------------------------------------|
| Fast and easy to implement                           | Assumes equal-sized, spherical clusters                |
| Interpretable and tunable                            | No soft membership (e.g., songs in multiple playlists) |
| Enhanced with feature weighting and size constraints |                                                        |

## Verdict:

With weighted features and constrained sizing, KMeans becomes a practical and human-aligned solution





# Next Steps

Use UMAP/t-SNE for visualizing clusters

Incorporate genre tags, lyrics, or user preferences

Explore soft clustering (e.g., GMM)





# Final Thought



## Beyond Data

**Clustering** isn't just about clustering data



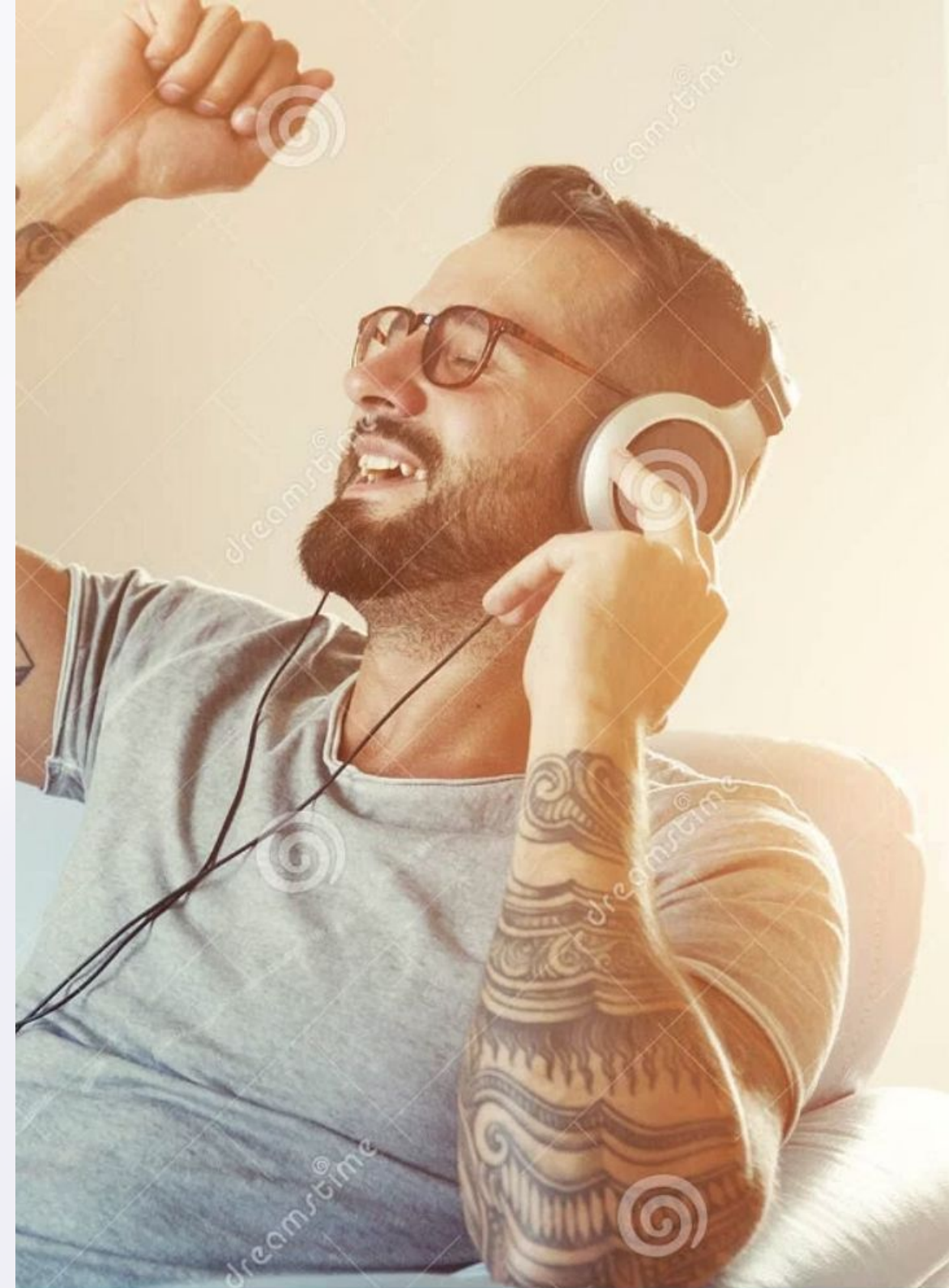
## Human Perception

They're about mimicking how humans perceive, group, and enjoy music

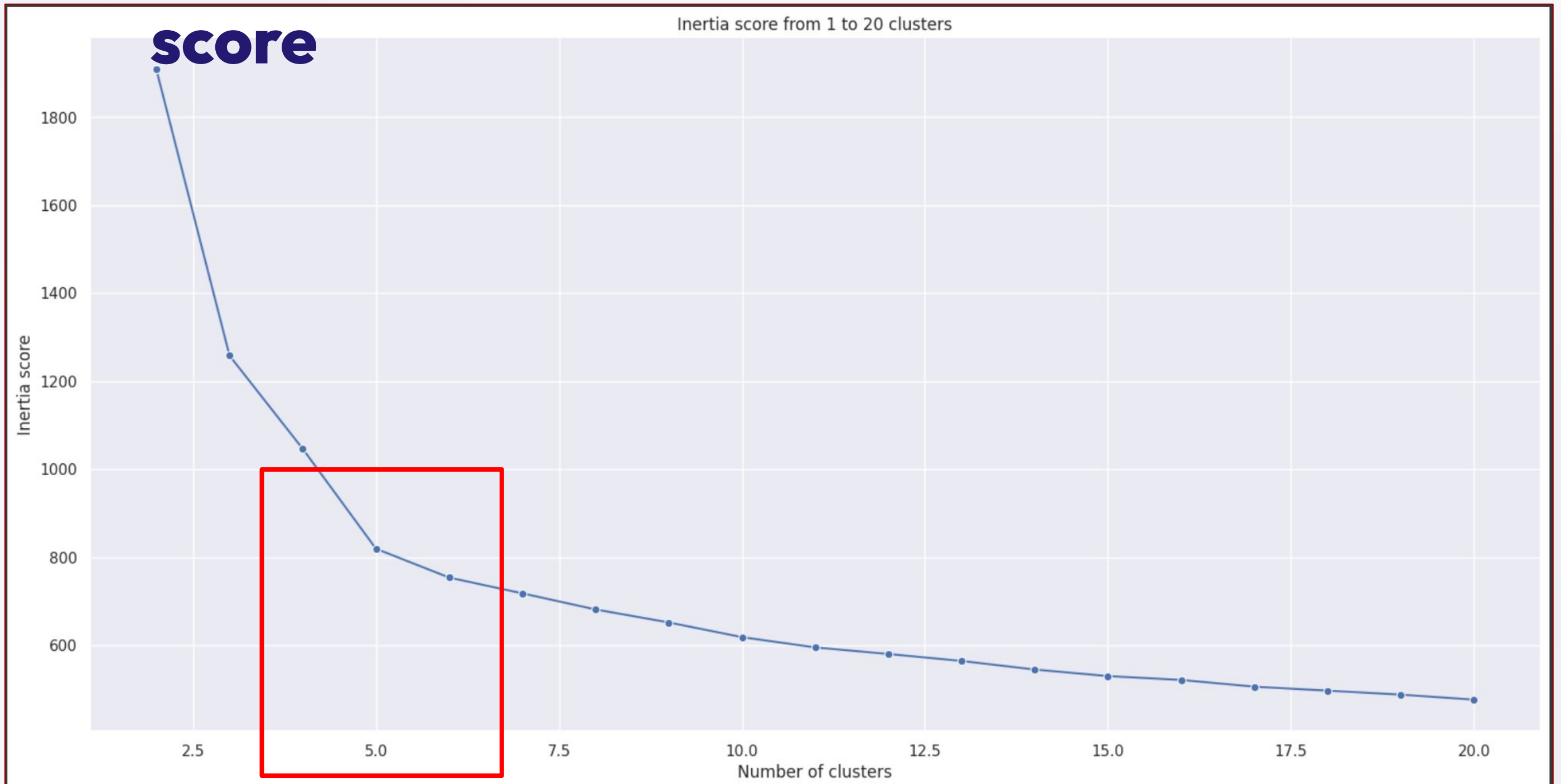


## Human-Like

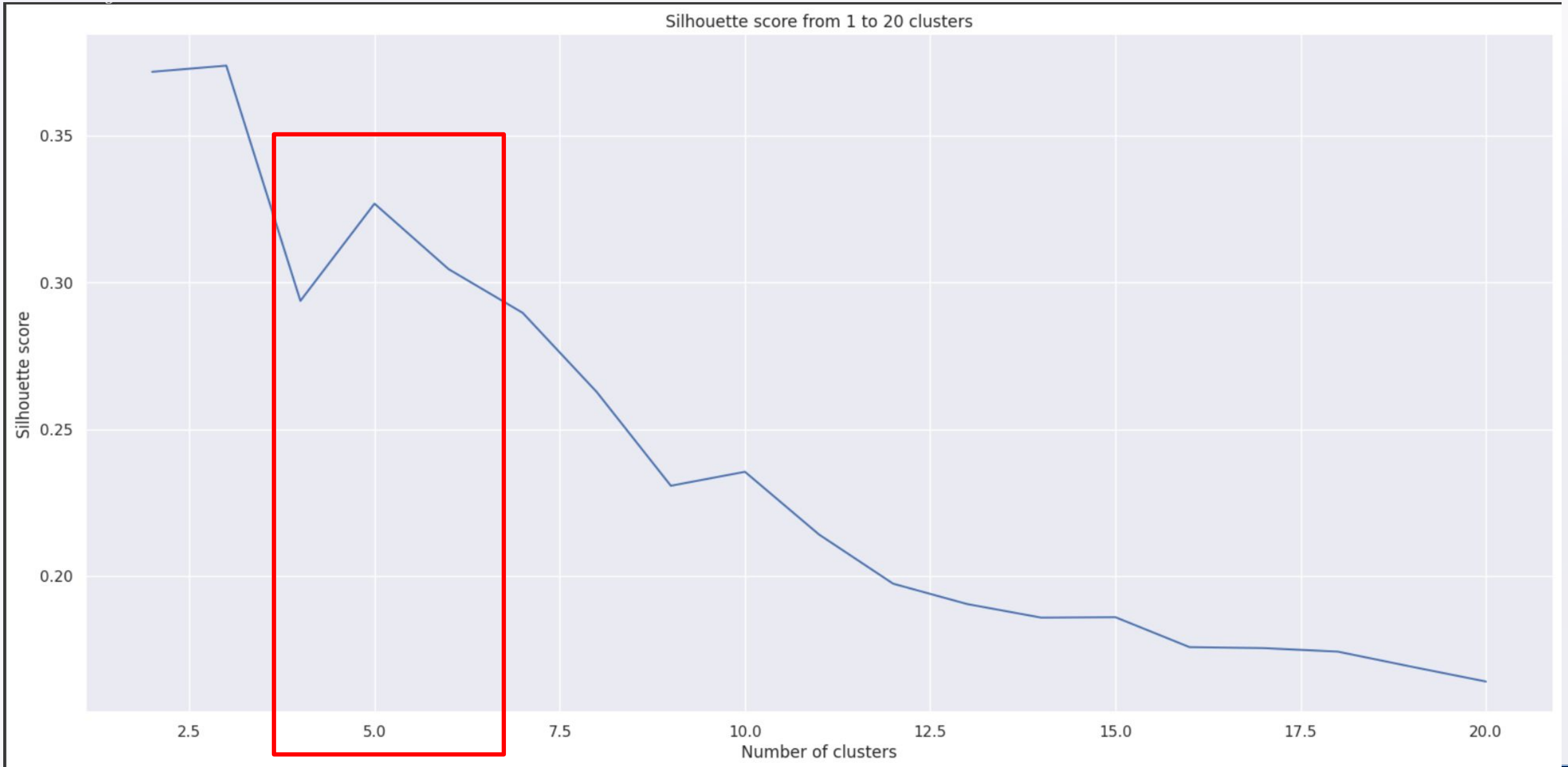
**Listening** algorithms helped us teach the algorithm to "listen like a human"



# First step of clustering: using Inertia



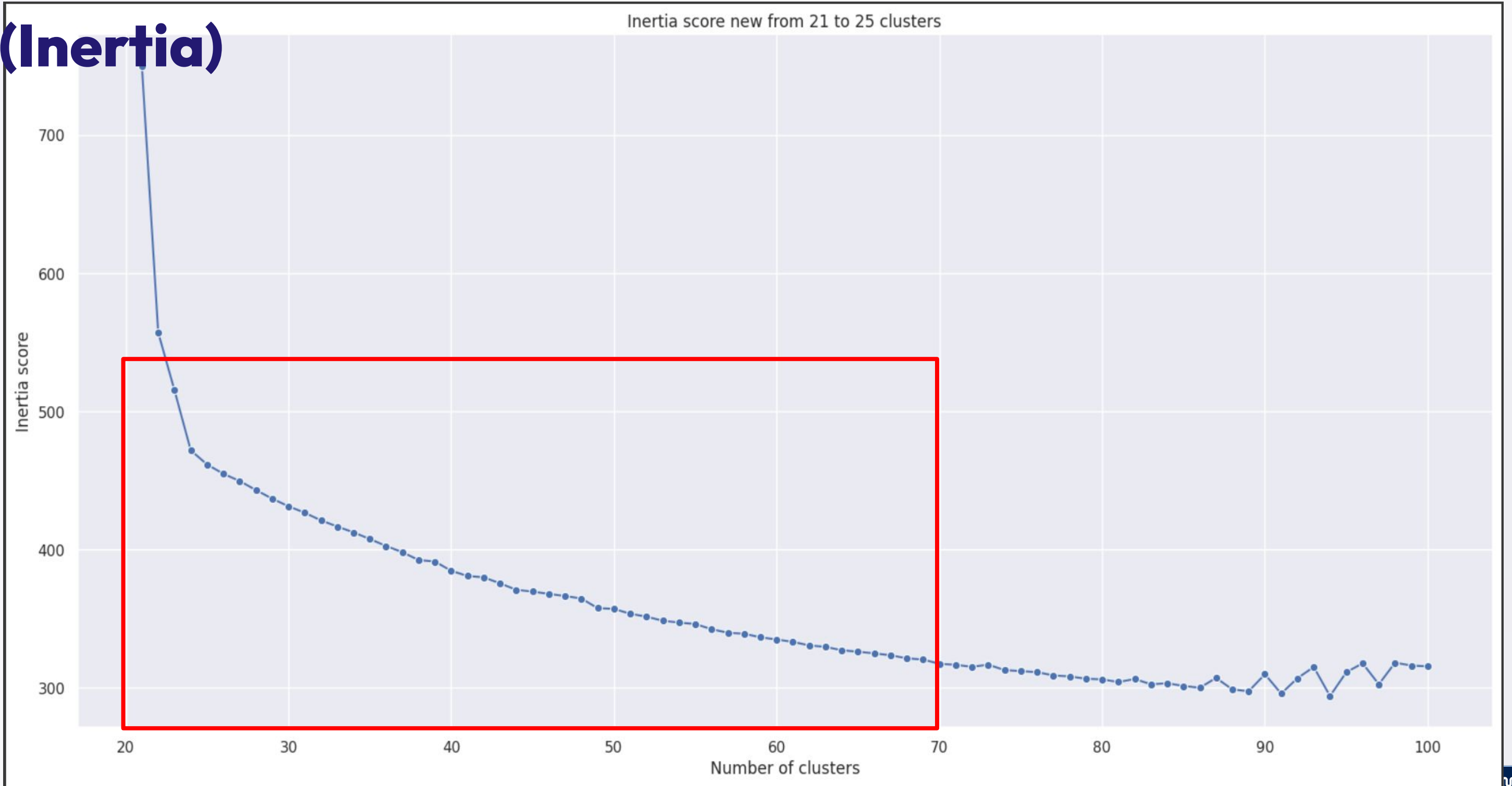
# First step of clustering: using Silhouette score



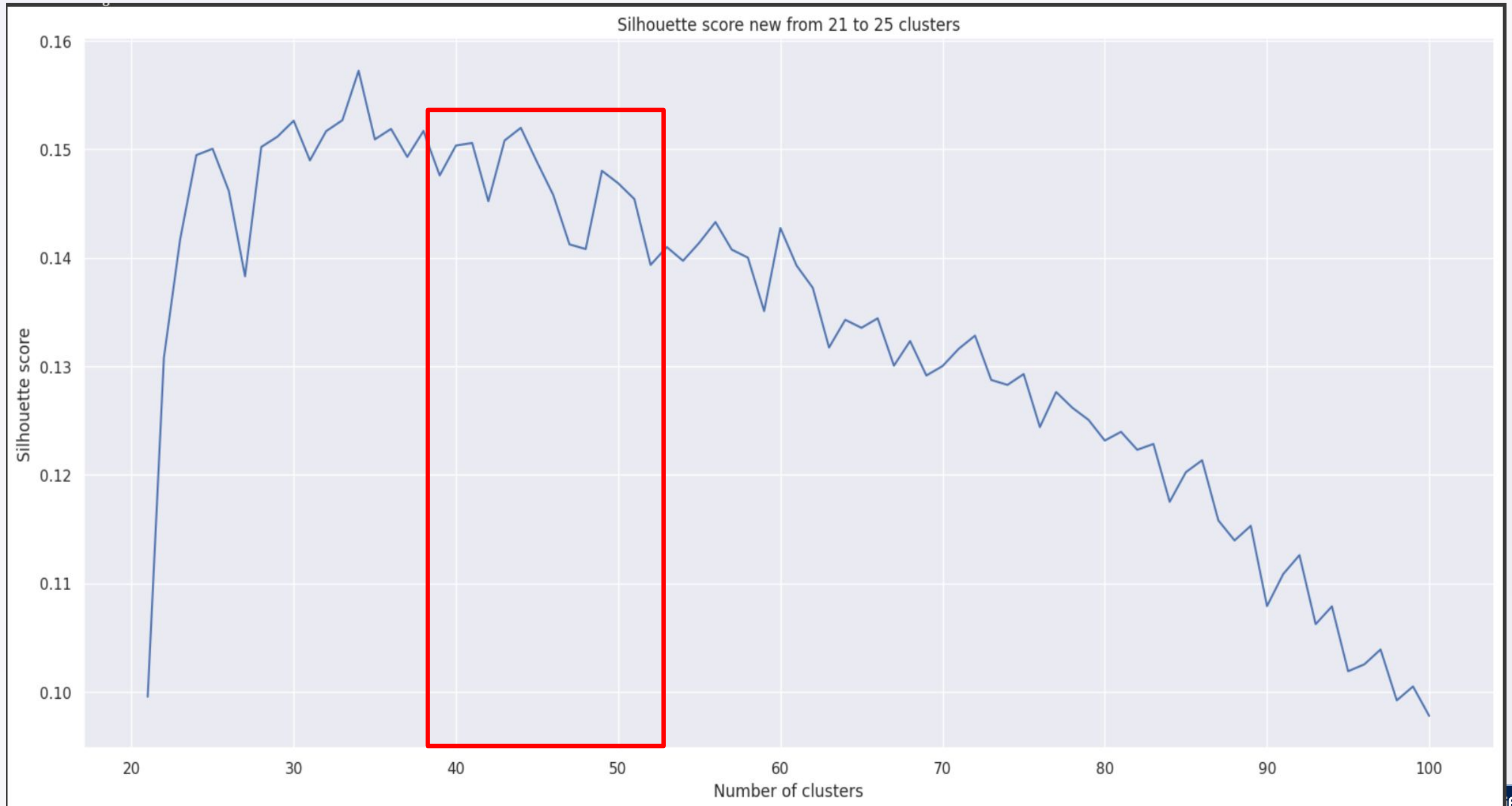


# Finding n of clusters for K-Means constrained

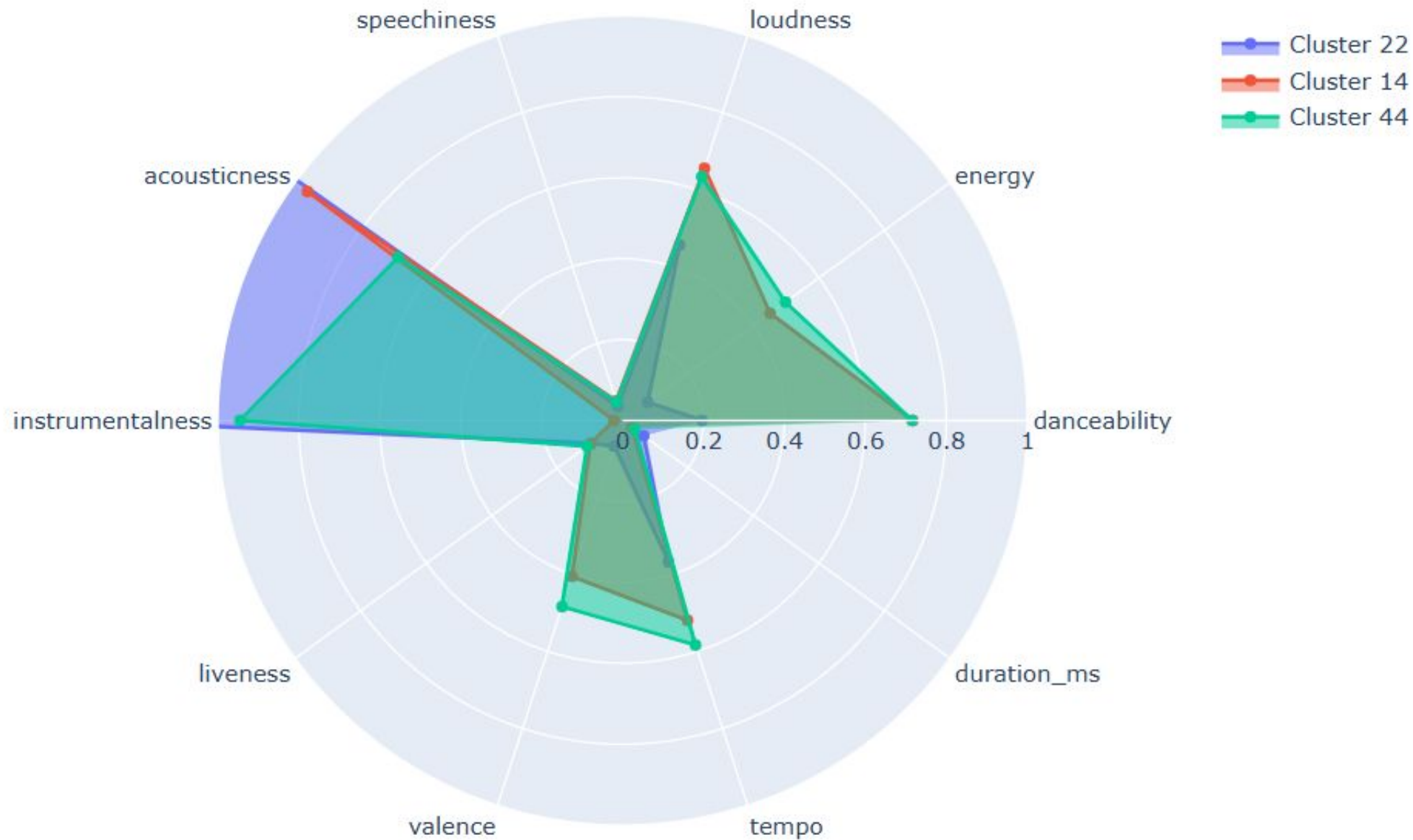
(Inertia)



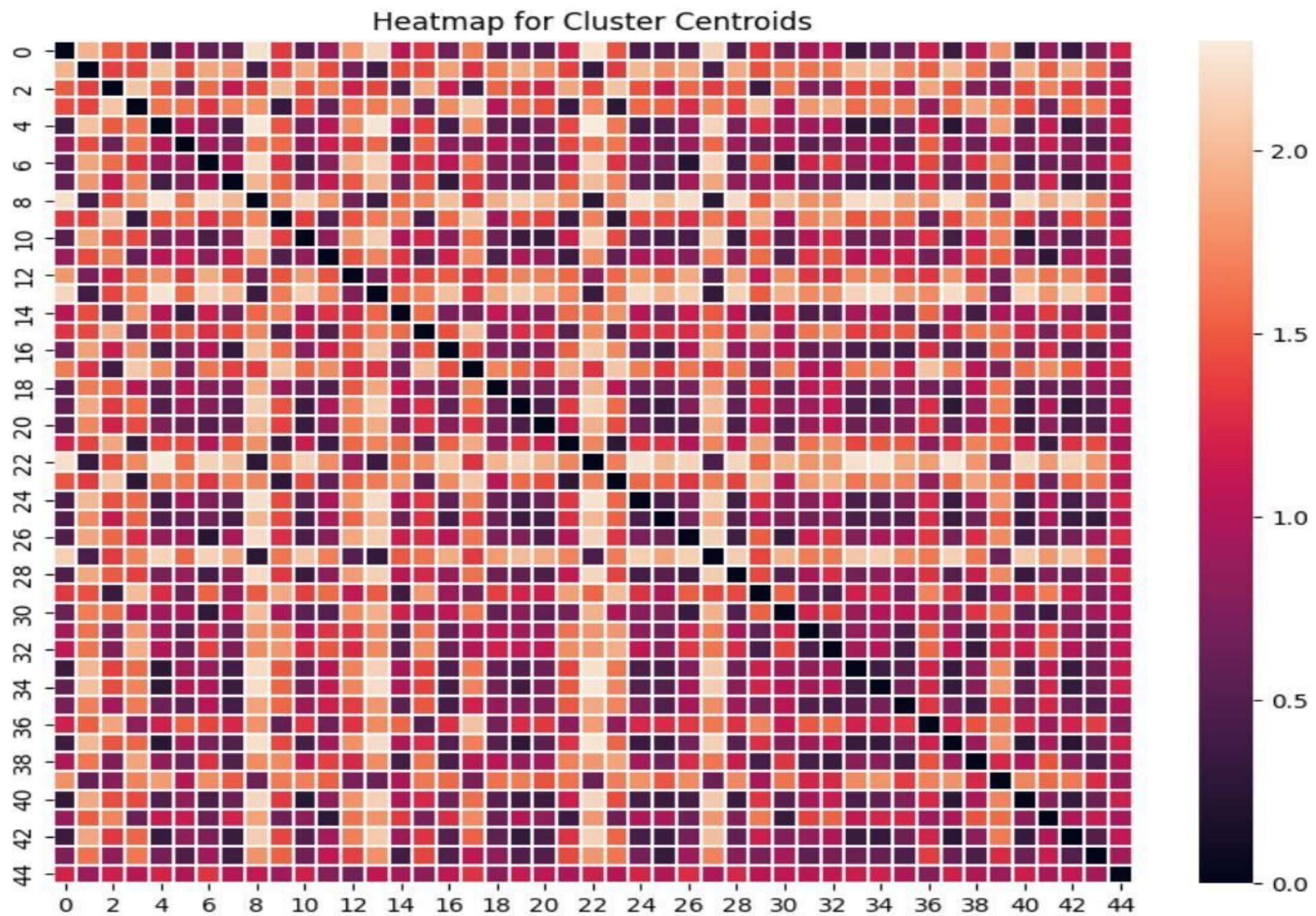
# Finding n of clusters for K-Means constrained (Silhouette)

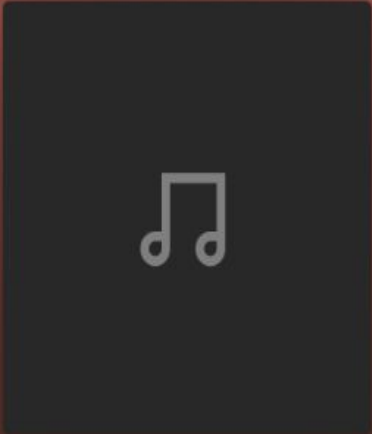


Radar chart of mean songs features by cluster









Public Playlist

# my\_playlist\_cluster\_1

this is a test list

Hassan syrian • 5 songs, 23 min 45 sec



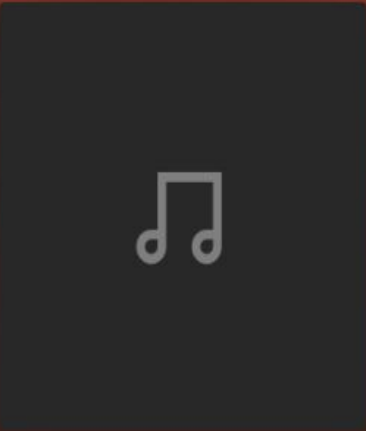
List

| # | Title                                                                                                                                                                                     | Album                        | Date added     |      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|------|
| 1 |  <b>Serenade To A Soul Sister - Remastered</b><br>Horace Silver                                         | Serenade To A Soul Sister    | 26 minutes ago | 6:19 |
| 2 |  <b>Warm Shadow</b><br>Fink                                                                            | Perfect Darkness             | 26 minutes ago | 5:46 |
| 3 |  <b>Karoliina Kaunis</b><br>Maria Gasolina                                                             | Mä Olen Sun                  | 26 minutes ago | 4:11 |
| 4 |  <b>Son of a Preacher Man</b><br>Aretha Franklin                                                       | This Girl's in Love with You | 26 minutes ago | 3:19 |
| 5 |  <b>Beauty and the Beast - from the Soundtrack "Beauty and the Beast"</b><br>Céline Dion, Peabo Bryson | Celine Dion                  | 26 minutes ago | 4:09 |

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Public Playlist

# my\_playlist\_cluster\_2

this is a test list

Hassan syrian • 5 songs, 33 min 55 sec

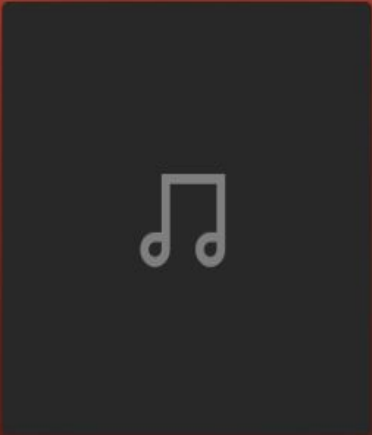
List

| # | Title                                                                                                                                     | Album                                                       | Date added     |       |
|---|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------|-------|
| 1 | <b>Yanagiwara</b><br>Hirotaka Shirotsubaki                                                                                                | Hyogo                                                       | 27 minutes ago | 7:27  |
| 2 | <b>unother</b><br>Terekke                                                                                                                 | Improvisational Loops                                       | 27 minutes ago | 1:02  |
| 3 | <b>Mussorgsky, Ravel: Pictures at an Exhibition, M. A 24: IV. Bydlo</b><br>Modest Mussorgsky, Oslo Philharmonic Orchestra, Mariss Jansons | Mussorgsky & Rimsky-Korsakov: Orchestral Works              | 27 minutes ago | 2:49  |
| 4 | <b>Choros No. 8</b><br>Heitor Villa-Lobos, Linda Bustani, Ilan Rechtman, Orquestra Sinfônica do Estado de S...                            | Villa-Lobos, H.: Choros, Vol. 2 - Choros Nos. 1, 4, 6, 8, 9 | 27 minutes ago | 20:17 |
| 5 | <b>Olancha Farewell</b><br>Harold Budd                                                                                                    | Lovely Thunder                                              | 27 minutes ago | 2:18  |

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Public Playlist

# my\_playlist\_cluster\_3

this is a test list

Hassan syrian • 5 songs, 27 min 17 sec

List

| # | Title                                                                                                                                       | Album                            | Date added     |      |
|---|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------|------|
| 1 |  <b>Be - Instrumental</b><br>Steve Angello, Laidback Luke | Be                               | 27 minutes ago | 6:49 |
| 2 |  <b>Sonnentanz</b><br>Klangkarussell                     | Netzwerk                         | 27 minutes ago | 6:00 |
| 3 |  <b>I Love You Always Forever</b><br>Donna Lewis         | Now in a Minute                  | 27 minutes ago | 3:59 |
| 4 |  <b>How to Disappear Completely</b><br>Radiohead         | Kid A                            | 27 minutes ago | 5:56 |
| 5 |  <b>Blue Bossa - Remastered</b><br>Joe Henderson         | Blue Note Trip 2: Sunset/Sunrise | 27 minutes ago | 4:31 |

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